Jeremy Burke

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Professional Summary

Software engineer with experience building and maintaining complex production software across simulation, computer vision, AI/ML, and web platforms. Strong in C++, Python, C#, TypeScript, distributed systems, and technical problem solving. Comfortable working in large codebases, collaborating across teams, integrating changes safely, and turning ambiguous requirements into reliable features, documentation, and user-facing tools.

Experience

Software Engineer — Sonalysts, Inc. *Waterford, CT* Aug 2026 – Present

- Develop and maintain a large-scale naval training and simulation product in a mature C++/Lua codebase, delivering defect fixes, platform upgrades, and new features through a collaborative multi-developer workflow
- Implement and troubleshoot distributed simulation capabilities, including DIS networking and simulation interoperability features such as IFF, while preserving compatibility across interconnected systems
- Use Qt Designer and related C++ tooling to improve operator-facing interfaces and support maintainable workflows in a complex, long-lived software system
- Write extensive technical documentation for software behavior, configuration, and user workflows, translating complex simulation capabilities into clear guidance for training-system users
- Contribute technical input to SBIR proposal development through software research, feasibility analysis, and concise technical writing

Computer Engineer — Beacon Industries *Newington, CT* Aug 2025 – Aug 2026

- Led SBIR/STTR technical programs across DoD, NASA, and MDA efforts, delivering reports, software prototypes, simulations, and technical analyses valued at $30K–$250K+ independently
- Developed computer vision models, synthetic sensor pipelines, and fusion-validation workflows for perception and multi-spectral detection programs
- Engineered custom plugins and simulation tools for thermal, environmental, and multi-spectral sensor modeling from raw data streams
- Built AI-enabled internal tools using AWS, Weaviate, OpenAI APIs, and custom automation pipelines for technical analysis and decision support
- Presented technical work, R&D outcomes, and commercialization opportunities to government stakeholders

Personal Projects

- **SentinelGrid — Climate-Risk Telemetry Platform:** End-to-end IoT pipeline — C++ sensor-fleet simulator, MQTT, FastAPI, PostgreSQL/PostGIS, and Python anomaly-scoring worker (z-score + Isolation Forest, sensor-drift quarantine) behind a Next.js/Leaflet national ops console (incident triage, playback, forecasts); deterministic in-browser simulation of 4,174 nodes across 19 US regions with zero-storage 24-hour replay, overlaid on live public data (NEXRAD radar, ∼3,700 NWS/USGS stations, active storm warnings); 90+ tests, 8 CI jobs.
- **ProjectPulse:** Production-style project-management platform — ASP.NET Core 8 Clean Architecture API with a React 19 dashboard; drag-and-drop Kanban with domain-enforced transitions, isolated demo workspaces, role-based permissions, rate limiting, and audit logging (xUnit/Vitest, Docker Compose, GitHub Actions CI).
- **Open Computer Vision Detection Dashboard:** Next.js/TypeScript + Cloud Run FastAPI dashboard serving 7 detectors across YOLO v8–v12, RT-DETR, and open-vocabulary YOLO-World; upload, webcam, and batch inference with annotated image, JSON, and CSV outputs.
- **Technical Paper AI Search Assistant:** Source-grounded RAG assistant on Cloudflare Workers/D1 — hybrid semantic + BM25 retrieval, in-browser PDF search (Transformers.js), exact-passage citations, and a fail-closed cost quota.

Education

University of Connecticut Aug 2019 – Dec 2023

B.S. Computer Science & Engineering, Minor Mathematics — GPA: 3.3 Storrs, CT

Technical Skills

- **Programming:** Python, TypeScript, JavaScript, C, C++, C#, Java, Lua, SQL
- **AI/ML:** YOLOv8, OpenCV, RAG, Sentence Transformers, LLM Systems, Multimodal Fusion
- **Web/Backend:** Next.js, React, FastAPI, ASP.NET Core 8, EF Core, REST APIs, Swagger/OpenAPI
- **Simulation/Systems:** DIS networking, IFF, Gazebo, ANSYS Workbench, MATLAB, Magics, Physics-Based Modeling
- **Cloud/Tools:** AWS, GCP/Cloud Run, Docker, Firebase, GitHub Actions, Qt Designer, Git, Linux, LaTeX